

Dylan Atkins

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Education

Florida State University, Tallahassee, FL

B.S. Chemical Engineering | Expected May 2030

GPA: 3.74/4.00

Palm Beach State College, Palm Beach Gardens/Lake Worth, FL

Associate of Arts | May 2026

Palm Beach Central High School, Wellington, FL

Academic Honor, AICE Recipient, Diploma | May 2026

Weighted GPA: 4.64/6.00

Key Science & Math Coursework: General Chemistry I & II + Labs (A), Calculus I (B), Calculus II (A), Calculus III (A)

Work & Research Experience

Shaw Industries – Plant 70, Bainbridge, GA |

August 2026

Chemistry Laboratory & Process Engineering Job Shadow

- Shadowed a Chemist and Process Engineering team to gain insight into industrial chemistry, manufacturing processes, and engineering career pathways Optimized crosslinking systems.
- Gained exposure to polymer testing, quality control, process optimization, and manufacturing operations, and how chemistry and engineering work together to improve product quality, efficiency, and production

Independent Research Project |

2025–2026

Optimization of Poly (acrylic acid) Polymer Networks for Extrusion and 3D Forming Applications

Supervisor(s): William Bartenslager (william.bartenslager@palmbeachschools.org)

- Developed low-temperature PAA–PVA hydrogel filaments as sustainable alternatives to petroleum-based 3D printing materials
- Optimized crosslinking systems (borax, glycerol, CaCl_2 , NaPAA), achieving significantly improved tear force and extensibility (t-test validated)
- Designed PAA–PAH polyelectrolyte outer films to mitigate shear-thinning and enable continuous extrusion with partial self-healing
- Conducted mechanical testing, solvent stability analysis, and extrusion flow diagnostics

Honors: First Place, School Science Fair (2025); Finalist, Palm Beach Regional Science Fair (2025)

Coastal Ocean Explorers: Shark Program |

(2025)

ANGARI Foundation & Florida International University

Supervisors: ANGARI Foundation; Florida International University

- Conducted a group biotechnology project to identify shark prey species found in local food markets
- Set up Baited Remote Underwater Video Systems (BRUVS) and drum lines in the Atlantic Ocean
- Collected fish specimens from multiple local markets
- Performed DNA extraction, PCR, gel electrophoresis, and submitted samples for gene sequencing to identify fish species • Used DNA Subway to create a phylogenetic tree and prepared a poster for presentation at an upcoming conference

Independent Research Project |

2024–2025

Effect of Aluminum Hydroxide on Lithium-Ion Separation

Supervisor(s): William Bartenslager

- Investigated lithium-ion separation from lithium carbonate using aluminum hydroxide precipitation under acidic and basic pH conditions
- Performed titrations, pH monitoring, mass analysis, and statistical validation using t-tests
- Demonstrated significant pH-dependent differences in lithium concentration and separation efficiency

Honors: First Place, School Science Fair (2024); Third Place Chemistry & Finalist, Palm Beach Regional Science Fair (2024)

Independent Research Project

2023–2024

Perfluorooctanoic Acid (PFOA) Removal from Water Using Dimethyl Sulfoxide and Granular Activated Carbon

Supervisor(s): William Bartenslager

- Designed a cost-effective remediation approach for PFAS contamination using solvent-assisted adsorption methods
- Conducted controlled GAC adsorption trials, mass comparison analyses, and optimization studies
- Evaluated environmental and health implications of PFOA contamination

Honors: First Place, School Science Fair (2023)

Technical Skills

Laboratory & Materials Science: Polymer synthesis; hydrogel fabrication and crosslinking; solution preparation; titration; pH measurement (LabQuest); mechanical testing; experimental design and optimization; statistical analysis (t-tests).

Programming & Data: Python (scientific computing, data analysis, automation fundamentals); Microsoft Excel; data visualization; experimental data analysis.

Engineering Design (CAD): Autodesk Inventor; Autodesk Revit; Onshape; Fusion 360; engineering drawings and 3D modeling.

Biotechnology: DNA extraction; PCR; gel electrophoresis; DNA Subway & BLAST analysis; sterile technique; microbial culturing (Kirby–Bauer method).

Awards & Honors

- QuestBridge National College Match Finalist, 2026
- Planting Science Star Project Awardee, 2025
- Palm Beach Regional Science Fair Finalist, 2025
- 1st Place, Palm Beach Central School Science Fair, 2023–2025
- 3rd Place, Chemistry, Palm Beach Regional Science Fair, 2024
- Citizen Science Participant, 2025
- Science Olympiad Participant, 2025
- Max Planck Brain Bee Participant, 2025